November 11, 2025

SYSTEM REQUIREMENTS **REPORT**

Prepared for Prof. May El Barachi

Presented By **TROJAN HEX**

REONA LOBO	8807206	ADALIA ALEX	8373759
TANIA BHATIA	84003474	FATEMAH ZAREEI	7612953
SUWATHI RAJASEKAR	8393606	MISHAIM JAT	8792501

Amendment history

Version	Date	Modified By	Reviewed By	Amendment details
1.0	28.10.2025	Trojan Hex	Prof. May El Barachi	Initial Version
1.1	11.11.2025	Trojan Hex	Kent PLC	Changed compliance framework from ISO 27001 to NIST

Table of Contents

1. Introduction 3
1.1 Purpose 3
1.2 Notation 3
1.3 Scope 3
1.4 Context Diagram 3
1.5 Definitions and Acronyms 4
1.6 References 4
1.7 Overview 4
2. General Description 4
2.1 System Functions 4
2.2 User Characteristics 5
3. General Constraints 6
3.1 Software Constraints 7
3.2 Hardware Constrains 7
4. Assumptions and Dependencies 7
4.1 Operational Assumptions 7
4.2 Environmental and Organization Assumptions 8
4.3 Technical Dependencies 8
5. Functional Requirements Master List 8
6. Functional Requirement 10
Use case overview: 11
6.1.1. Create Account 11
6.1.2. Log In 12
6.1.3. Sign Out 13
6.2. Manage Users 14
6.3. Integrate External Regulatory Frameworks 15

6.4.1. Upload Compliance Documents 16
6.4.2. Analyze Document for Compliance Gaps (NLP & AI Engine)..... 17
6.4.3. AI Suggestion Engine 18
6.5. Track Compliance Progress and Status via Dashboard 19
6.6.1. Invite Collaborators for Collaboration..... 20
6.6.2. Allows Multiple Users to Comment 21
6.6.3. Allows Users To Edit Shared Files..... 22
6.7. Notify Compliance Officer About New or Updated Requirements..... 23
6.8.1. Query AI chatbot for Compliance related answers..... 24
6.8.2. User provide feedback to AI chatbot to improve accuracy over time 25
6.9.1. Generate Audit Ready Compliance Reports 26
6.9.2. Maintain Audit Trails and Logs..... 27
6.10. Secure Data Storage and Encryption 28
7. External Interface Requirements..... 30
7.1 Data Interface..... 30
7.2 User Interface..... 31
7.3 Other Interface 32
8. Non-Functional Requirements 32
8.1 System Performance 32
8.2 Information security..... 33
8.3 Availability 33
8.4 Capacity 34

1. Introduction

1.1 Purpose

ComplyAI is an IT compliance management system that develops the process of reviewing, validating, and tracking compliance of organizations with various standards, including NIST, ITIL, and HIPAA. The system assists compliance officers and administrators to simplify the audit preparation with the aid of NLP and AI-based analysis to identify non-compliant areas, recommends corrections, and creates audit-ready reports in a strictly controlled and interactive atmosphere.

1.2 Notation

All the diagrams and models in this paper are created on the Unified Modeling Language (UML) notation and the UML is used to show the interactions and the workflow in the system through the use case diagrams, activity diagrams and system context diagrams.

1.3 Scope

The ComplyAI is an intelligent and centralized system to keep the IT in check. Its main objectives include:

- Posting and storing of compliance documents.
- There is the automatic analysis of documents against regulatory frameworks via NIST and ITIL.
- Real-time monitoring of compliance by an interactive dashboard.
- Generating reports and AI recommendations for compliance improvement.

The platform is to improve the level of efficiency, manual review attempts, and improve the accuracy of auditing in both small and large organizations.

1.4 Context Diagram

- ComplyAI system communicates with various actors and external structures as shown in the context diagram:
- Compliance Officer: Uploads and maintains compliance documents, reviews AI suggestions, and monitors progress.
- System Administrator: Responsible of user management, access control, and data security.
- ComplyAI System: Documents parsing, compliance validation and reporting.
- External Structures: Provide regulatory updates and compliance resources (e.g., NIST and ITIL libraries).

1.5 Definitions and Acronyms

- AI: Artificial Intelligence
- NLP: Natural language processing.
- NIST: National Institute of Standards and Technology
- HIPAA: Health Insurance Portability and Accountability Act.
- ITIL: Information Technology Infrastructure Library.

1.6 References

- ComplyAI Team (2025) *ComplyAI Planning and Feasibility Report*. University Capstone Project Document.
- ComplyAI Team (2025) *System Architecture Diagram*. Internal project design artifact.
- National Institute of Standards and Technology (NIST) (2018) *Framework for Improving Critical Infrastructure Cybersecurity*, Version 1.1. Gaithersburg, MD: U.S. Department of Commerce.
- AXELOS (2019) *ITIL® Foundation: ITIL 4 Edition*. London: TSO (The Stationery Office).
- Office of the National Coordinator for Health Information Technology (ONC) (2013) *Health Insurance Portability and Accountability Act (HIPAA) Privacy, Security, and Breach Notification Rules*. U.S. Department of Health and Human Services.

1.7 Overview

ComplyAI is a single compliance management space that uses AI to conduct document analysis, compliance tracking, and audit readiness.

It promotes precision, openness, and cooperation in the administration of IT criteria, which allows organizations to hold similar adherence to several frameworks.

2. General Description

2.1 System Functions

- **User Account Management**
  - **Create Account:** Allows new users to register by providing details such as username, email, and password. User credentials are securely stored using encrypted authentication.
  - **Login:** Authenticates registered users through secure credentials, granting access to system functionalities based on user role.

- **Sign Out:** Enables users to log out safely, terminating active sessions and clearing session tokens.
- **Manage Users:** Administrators can view, update, or assign user roles (e.g., compliance officer, reviewer, admin).
- **Remove Users:** Admins can delete inactive or unauthorized users while maintaining audit trail logs.

- **Analyze Document for Compliance Gaps**
  - Users upload a document and select a supported compliance framework (e.g., ISO 27001, GDPR, HIPAA, NIST).
  - The NLP engine scans the document to identify non-compliant, missing, or weakly aligned sections.
  - Detected gaps are highlighted, and the AI provides rewrites aligned with the selected standard.
  - The system learns continuously from uploaded data, user feedback, and updated compliance rule, improving its understanding and accuracy over time.

- **AI Suggestion and Auto-Fix**
  - Provides context-aware rewriting of non-compliant text.
  - Users can choose to automatically apply the AI's corrections

- **Collaboration and Dashboard Analytics**
  - Supports real-time document co-editing and annotation.
  - Displays visual analytics such as compliance score, gap summaries, and progress tracking.
  - Provides an activity timeline and version control to ensure audit transparency.

- **Security, Access Control, and Logging**
  - Implements Role-Based Access Control (RBAC) to manage user permissions.
  - Maintains encrypted audit logs of all document edits and AI interactions.
  - Ensures data integrity through secure communication, encryption, and cloud-based backups.

2.2 User Characteristics

- **Compliance Officer (Primary User):**
  - Background: Compliance professionals responsible for preparing and reviewing policy documents.
  - Main Functions: Upload documents, run compliance checks, review AI-generated suggestions, and approve corrections.
  - Interface Type: Self-service, document upload, and dashboard view.

- Training Level: Moderate; requires familiarity with compliance frameworks (ISO 27001, GDPR, etc.).

- **Administrator (System Manager):**

  - Main Functions: Manage user accounts, roles, and system settings; upload or update compliance frameworks.
  - Interface Type: Administrative dashboard with role-based access control.
  - Training Level: Trained user with knowledge of system management and compliance processes.

- **AI Model Trainer / Data Engineer (Technical User):**

  - Background: Technical staff responsible for maintaining, retraining, and updating AI and NLP models.

  - Main Functions: Feed new compliance data, monitor AI performance, and manage the learning database.

  - Interface Type: Backend data interface and analytics dashboard.

  - Training Level: Advanced; requires knowledge of NLP, ML, and data pipelines.

Predicted User Platforms:

- **Devices:** Desktop and Laptop (primary), Tablet (optional)

- **Operating Systems:** Windows 10/11, macOS Monterey+, Ubuntu 22.04+

- **Web Browsers:** Google Chrome, Microsoft Edge, Safari, Firefox

- **Access Mode:** Web-based interface (React.js frontend) with secure backend API connectivity

- **Connectivity:** Requires stable internet access for document uploads, real-time collaboration, and AI inference.

3. General Constraints

The project encounters various constraints because of the evolving nature of the compliance standards, the technological foundation required for the AI-powered analysis and the constant need for human oversight to guarantee accuracy. Other constraints include difficulties with

merging legal systems, supporting scalability and ensuring robust privacy of data and security for confidential compliance documents.

3.1 Software Constraints

- Advanced AI and NLP models trained on a particular range of IT compliance standards and production documents must be used by our solution.
- A python based backend (Flask) and an intuitive frontend (Node.js, React.js) is needed.
- The system depends on libraries and frameworks like Hugging Face, Transformers and relies on APIs (Rest, GraphQL), and leading RAG orchestration tools (Langchain).
- Document storage and vector databases (PostgresSQL, MongoDB) are important.
- Following encryption standards (TLS/SSL), Safe real-time collaboration (Websockets), deployment with Docker/Kubernetes and Integration with version control is essential.
- The software should also integrate several testing frameworks (PyTest, Selenium), and CI/CD pipelines (GitHub.)
- Integration with company specific platforms is mandatory for extensibility and quick updates are mandatory to align with changing compliance requirements.

3.2 Hardware Constrains

- Intel i5 10th Gen, Ryzen 5 or Apple M1 processor are the minimum hardware requirements.
- Minimum of 16 GB RAM is necessary to easily run containers, browsers and databases.
- Minimum of 512 GB SSD of storage is required to handle environments, source code and cache Docker images.
- Encouraged operating systems include macOD Monterey, Ubuntu 22.04 LTS, Windows 11 Pro.

4. Assumptions and Dependencies

4.1 Operational Assumptions

- Users will have a stable internet connection to enable real-time collaboration, AI inference, and document uploads.
- The system will be deployed either on-premise or cloud-based (AWS/Azure) depending on organizational data policies and privacy requirements.
- AI-generated recommendations will require human validation before being finalized or approved for audit submission.
- Regulatory and compliance standards (ISO 27001, NIST, ITIL, HIPAA, PCI DSS, etc.) will be accessible via official data sources or updated repositories to ensure accurate validation.
- The organization will allocate compliance experts to periodically review and validate AI outputs to ensure reliability and compliance integrity.

4.2 Environmental and Organization Assumptions

- Compliance officers, administrators, and engineers will have the necessary training and awar eness to interact with compliance frameworks and AI-driven tools.
- The organization will provide domain-specific compliance documents to fine-tune and customize the internal AI copilot.
- Regular system updates and retraining of AI models will be conducted to stay aligned with evolving compliance standards.
- The platform will be operated within environments that meet enterprise-grade security and regulatory standards, ensuring safe handling of sensitive data.

4.3 Technical Dependencies

- The system depends on AI/NLP frameworks such as Hugging Face Transformers, spaCy, and LangChain/LlamaIndex for document parsing, clause mapping, and context-aware analysis.
- The backend relies on Docker for containerization and Kubernetes for scalable deployment.
- Cloud storage services (AWS S3 or Azure Blob) are used for document storage and retrieval, while PostgreSQL and MongoDB handle structured and unstructured data.
- Security dependencies include TLS/SSL for secure data transmission, AES-256 for encryption at rest, and OAuth 2.0 / JWT for authentication and role-based access control.
- Integration with version control (GitHub/GitLab) and CI/CD pipelines is required for automated deployment and continuous testing.

5. Functional Requirements Master List

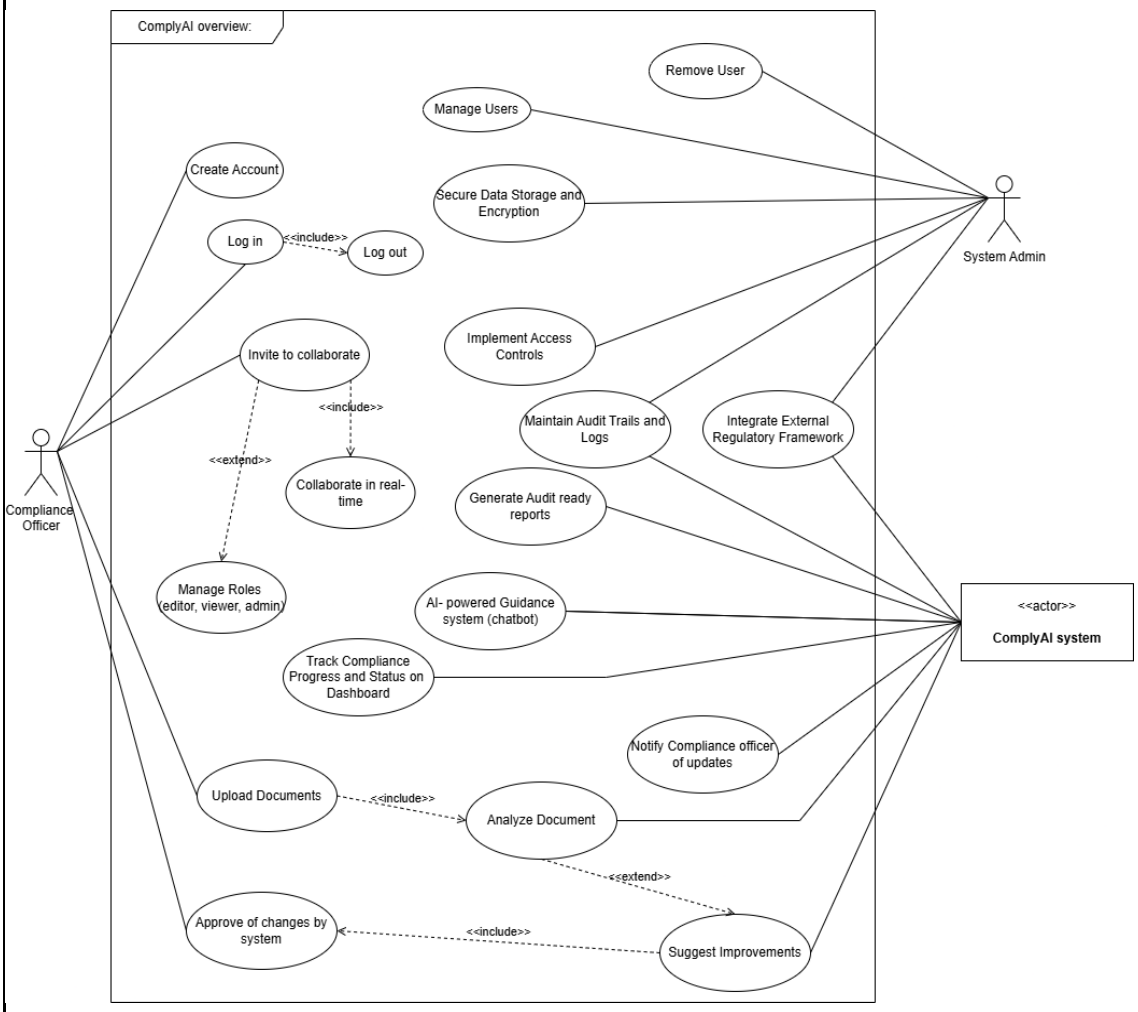
REQ ID	Requirement Name	Requirement Description
REQ - 1.1	Create Account	Enables a new user to register and create an account in the ComplyAI system by providing required details such as name, email, password, and organization. The system validates and stores the

		information securely, allowing the user to access the platform.
REQ - 1.2	Log In	Authenticates an existing user by verifying their credentials (email and password). On successful verification, the user gains access to their personalized dashboard and features according to their assigned role.
REQ - 1.3	Sign Out	Logs out the authenticated user, ending their active session securely by invalidating session tokens and redirecting them to the login or sign-up page to maintain account security.
REQ - 2	Manage Users	Allows the System Administrator to view, edit, or remove user accounts within ComplyAI. This includes assigning roles, updating permissions, and managing collaboration access for compliance projects
REQ - 3	Integrate External Regulatory Frameworks	The system can import and incorporate external regulation into the compliance validation and reporting pipeline to ensure the AI engine can validate documents against external frameworks as well.
REQ - 4.1	Upload Compliance Documents	Allows the compliance officer to upload IT compliance documents into the system for AI-based review. The system therefore stores and prepares files to be analyzed automatically.
REQ - 4.2	Analyze Document for Compliance Gaps (NLP & AI Engine)	Enables users to upload a document and select a compliance standard for automated analysis. The system uses NLP and AI to identify non-compliant sections, highlight them, and optionally apply AI-driven corrections to generate an improved, compliant version of the document.
REQ - 4.3	AI Suggestion Engine	It automatically detects non compliant clauses in the documents and suggests corrections which align to selected framework
REQ - 5	Track Compliance Progress and Status via Dashboard	Enables users to view compliance analysis results and progress in real time through the

		dashboard showing compliance percentages and document status.
REQ - 6	Collaborate in Real-Time on Documents	Allows several users to collaborate to create, edit and review files live by dealing with invitations, commenting and editing with role based authorization and immediate synchronization among users.
REQ - 7	Notify Compliance Officers of Regulatory Updates	Alerts compliance officers automatically of new or changed regulatory requirements and delivers AI-generated suggestions for changes to make sure the files meet the current standards.
REQ - 8	Provide AI-Powered Compliance Guidance (Chatbot)	Delivers on-demand, AI generated compliance support by letting users query regulations, receive custom/tailored answers, and submit the feedback, changes or corrections to consistently upgrade the chatbot's precision and relevance.
REQ - 9.1	Generate Audit Ready Complaint Report	Generates structured reports which summarize detected issues and corrective actions taken and provides the option to download the reports
REQ - 9.2	Maintain Audit Trails and Logs	Must automatically capture and store immutable logs of all user and system activities, including document actions and AI operations. Logs are timestamped, securely stored, searchable and accessible only to authorized roles for audition and compliance verification
REQ - 10	Secure Data Storage and Encryption	All Data, including documents, user credentials and AI assets, must be encrypted using TLS/SSL for transmission and AES-256 for storage. Encryption keys are securely managed and rotated regularly to ensure confidentiality, integrity and compliance with data protection standards.

6. Functional Requirement

Use case overview:



6.1.1. Create Account

Description:

- Allows a new user to register and create an account in the ComplyAI system.

System Input:

- User-provided registration information including name, email, password, and organization.

Display:

- The system displays the registration form with required fields and submission options.

System Processing:

- Validates user input, checks for duplicates, and creates a new account if data is valid.

System Output:

- Confirmation of successful account creation and redirection to the login page.

Constraints:

- All user data is securely stored and validated for correctness.

Data Handling:

- User information is stored in the ComplyAI database.

Error Handling:

- Handles duplicate email, missing or invalid data, and service unavailability by notifying the user.

6.1.2. Log In

Description:

- Authenticates a registered user and grants access to their dashboard.

System Input:

- User-provided email and password.

Display:

- Login page prompting for credentials.

System Processing:

- Verifies credentials against the database and grants access if correct.

System Output:

- Access to the dashboard upon successful authentication.

Constraints:

- Authentication data is securely processed.

Data Handling:

- Credentials checked against stored account information.

Error Handling:

- Handles invalid credentials, inactive accounts, and service errors.

6.1.3. Sign Out

Description:

- Securely logs out the user and ends the current session.

System Input:

- User-initiated log out action.

Display:

- Option to log out in the dashboard or menu.

System Processing:

- Invalidates the session and clears authentication tokens.

System Output:

- Redirection to the login page or home page.

Constraints:

- Session termination is secure.

Data Handling:

- Clears session data and authentication tokens.

Error Handling:

- Handles session timeout and logout errors.

6.2. Manage Users

Description:

- Allows the system admin to view, edit, or remove other users' access to documents in ComplyAI.

System Input:

- Admin selects a user and specifies permissions or removal.

Display:

- List of all invited users with options to edit roles or remove users.

System Processing:

- Updates user roles, permissions, or removes users in the system database.

System Output:

- Confirmation of changes made and updated user list.

Constraints:

- Admin privileges are required to perform actions.

Data Handling:

- Updates and saves user roles and permissions in the database.

Error Handling:

- Handles unauthorized access and database errors by notifying the admin.

6.3. Integrate External Regulatory Frameworks

Description:

The system can import and incorporate external regulation into the compliance validation and reporting pipeline to ensure the AI engine can validate documents against external frameworks as well.

System Input:

- External framework files or templates
- URL to official regulatory framework documentation

Display:

- List of available internal and external frameworks to choose from
- Status indicators which show which frameworks are successfully integrated

System Processing:

1 Format of the external framework file or URL is validated

2. Framework rules, clauses and requirements are parsed

3. Store framework metadata and rules for future use

4. Update the AI engine to include addition of new framework

System Output:

- Confirmation message shown to user of successful integration.
- Framework added to the list of existing frameworks in the system

Constraints:

- External frameworks must be in supported formats
- Only admins can integrate external frameworks into the system

Data Handling:

- Framework files stored securely
- Metadata logged with timestamp and user ID

Error Handling:

- If file format is not supported, displays error and rejects integration of the framework
- If parsing fails, user is notified
- If framework already exists user is asked to skip or overwrite
- Deny integration action to unauthorised users.

6.4.1. Upload Compliance Documents

Description:

Allows the compliance officer to upload IT compliance documents into the system for AI-based review. The system therefore stores and prepares files to be analyzed automatically.

System input:

- Authenticated user credentials
- Documents (.pdf, .docx, or .txt)
- Optional data (document name etc.)

Display:

- Upload file with plus icon or drag-and-drop.
- Success or error notifications.
- List of uploaded documents

System Processing:

- Validate file permissions.
- Encrypt and store documents securely.
- Extraction of compliance and mapping via AI.

System Output:

- Confirmation message
- Document displayed in dashboard, showing its current status.

Other:

- Auto-save function during interruptions.
- File size limit.

Constraints:

- Only authorized users can upload.
- File types restricted to PDF, DOCX, and TXT.

Data Handling:

- Document data encrypted using AES-256.
- Time-stamped logs maintained for updates.

Error Handling:

- Invalid file format ("please upload PDF,DOCX OR TXT")
- Permission error ("access denied. Contact admin for access")
- Duplicate upload ("file already exists")

6.4.2. Analyze Document for Compliance Gaps (NLP & AI Engine)

Description:
Automatically analyzes a document against a selected compliance standard and identifies and highlights non-compliant sections. Users can view the highlighted section and apply AI-based corrections.

System Input:

- Uploaded document and select a compliance standard.

Display:

- Options to upload documents, select standards, view non-compliant sections, and apply AI corrections.

System Processing:

- NLP & AI engine analyzes document, highlights gaps, and applies corrections if permitted.

System Output:

- AI-corrected document available for download or viewing.

Constraints:

- Processing must ensure document confidentiality.

Data Handling:

- Logs analysis results and stores temporary document copies securely.

6.4.3. AI Suggestion Engine

Description:

It automatically detects non compliant clauses in the documents and suggests corrections which

align to selected framework

System Input:

- Upload compliance document to be checked
- Select the compliance/framework standard

Display:

- Highlight non compliant sections
- Suggested corrections shown
- Confidence score for each correction shown

System Processing:

1. Clauses are parsed and extracted using NLP
2. Clauses are matched and compared against framework standards
3. Gaps inconsistencies and missing information are identified
4. AI generate suggested rewrites

System Output:

1. Non compliant sections are flagged
2. Correct suggestions generated by AI
3. Confidence scores of suggested rewrites
4. Updated document after user approval

Constraints:

- Only documents in PDF, DOCS, TXT formats are supported

- AI might be limited by the model used, human verification may be needed
- Maximum document size: 50 MB

Data Handling:

- Documents are stored for a period of 90 days
- Changes made to the document after suggestions are stored
- Actions of users are logged by audit trails

Error Handling:

- If the document is corrupt or in the wrong format, user must be asked to re upload
- If the confidence score is low, notify user for manual review and verification
- Deny access to unauthorized users

6.5. Track Compliance Progress and Status via Dashboard

Description:

Enables users to view compliance analysis results and progress in real time through the dashboard showing compliance percentages and document status.

System input:

- Authenticated User credentials.
- Compliance data from analyzed documents.

Display:

- Status of each uploaded document (pending, completed)
- Filters for document category, date etc.
- Dashboard showing compliance percentage per framework.

System Processing:

- Calculate overall compliance scores.
- Aggregate compliance data.
- Update data via APIs or WebSockets.

System Output:

- Dashboard view
- Report with compliance percentage

Other:

- Users can export compliance reports for audits.
- Admin can monitor compliance across departments.

Constraints:

- Only authorized users can access
- Data updates depended on server connection.

Data Handling:

- Data stored securely in a compliance database.
- Time-stamped to track updates.

Error Handling:

- Database connection failure ("unable to load dashboard")
- Unauthorized access ("access denied. Contact admin for access")

6.6.1. Invite Collaborators for Collaboration

Description:

This feature provides a system to invite collaborators to documents, permission for viewing or editing in real time and assigning roles.

System Input:

- Email ID or user IDs provided by document owners or admins to send invites with permissions assigned.

Display:

- Invitation status is visible to the admin and owners (accepted, declined, pending). Current collaborators and their roles are shown in the access control lists.

System Processing:

- Forwards invitation notifications, upon entry user identities are verified and permissions and access rights are updated inside the framework.

System Output:

- Users are notified of invitations and validation of access changes. Permissions and collaborator views are updated dynamically.

Other Constraints:

- Invitation workflow must follow security protocols which includes authorization standards and authentication.

Data Handling:

- Metadata of invitations are stored (roles, timestamps, invitee, inviter) and requires access through session confirmation and secure tokens.

Error Handling:

- Deals with invalid email IDs or withdrawn invitations with error messages. Unauthorized access attempts are blocked by failed authentication.

6.6.2. Allows Multiple Users to Comment

Description:
This feature supports multiple users to simultaneously comment on the shared files in real time, approving collaborative and contextual feedback.

System Input:

- Users enter comments attached to particular sections of the file through a shared environment.

Display:

- The framework exhibits comments along with the relevant file text dynamically with the user recognition and timestamps. Comments can be filtered by date, author or active/resolved status.

System Processing:

- The framework syncs incoming comments in real time, retains comment threads and user's views are updated immediately. It carries out role-based permissions on who can or resolve comments.

System Output:

- Notification for new comments and real time updated comments were shown to all collaborators with status indicators.

Other Constraints:

- Concurrent users must be supported without delays and support consistency throughout different platforms and client devices.

Data Handling:

- The input of text supports Unicode. Comments are stored along with its metadata that includes the linked file location, timestamp and the author.

Error Handling:

- Clashes from live changes in comments will trigger alerts from manual clarification. Network disturbances are handled by caching unsent comments and trying again until the synchronization is successful.

6.6.3. Allows Users To Edit Shared Files

Description:

- This feature permits users to edit shared files in real time with immediate synchronization of the changes made and also the conflict resolution mechanism.

System Input:

- Text edits, changes in formatting and structural notification are entered by the users through a collaborative editing environment.

Display:

- The file content updates real-time for all the users. Visible signals indicate who is editing which section. Version controls and editing history are all available.

System Processing:

- Real time text changes are processed using operational transformation or conflict free replicated data types (CRDT) to resolve simultaneous edits. Applying layout and formatting rules.

System Output:

- A continuously synchronized file displayed to all collaborators with maintained edit history and version snapshots that are automated.

Other Constraints:

- Editing requirements should be responsive even with high user concurrency. Role based access control limits the editing rights.

Data Handling:

- Encourages Unicode and common file formats. Edits are tracked with change type, author ID and timestamps.

Error Handling:

- Solves syncing conflicts automatically where it's possible, or notifies users and maintains conflicting versions for review.

6.7. Notify Compliance Officer About New or Updated Requirements

Description:
This feature alerts the compliance officers about new or updated requirements automatically, delivering them with AI-generated document modifying suggestions to make sure of the ongoing compliance.

System Input:

- The system accepts regulatory update feeds and scans internal compliance files. The AI model produces suggested text revisions or add-ons based on the changes detected.

Display:

- The compliance engineers are given live notifications and access review panels listing suggested changes, with the choice to filter by date, file section or regulation.

System Processing:

- The system spots regulatory updates, implements AI analysis to impacted files and creates suggested changes. It maintains user notifications and tracks approval status and review.

System Output:

- Alerts to compliance engineers, documented suggestions for changes and approval modifications reflected in updated compliance, all are included in the outputs.

Other Constraints:

- The system should integrate with several AI engines and regulatory update sources, makes sure of safe and secure data handling and support regular audit trails for all the rejections and approvals.

Data Handling:

- The input data comprises structured and unstructured regulatory text. Outputs adhere to standardized file format. Data approval ensures suggestions are relevant contextually and accurately linked to file segments.

Error Handling:

- Manages the failure in update feeds with retry mechanism. Alerts the ambiguous suggested or inconsistent changes for the officer's review. Alert delivery failures trigger alert notifications to the officers for manual follow-up.

6.8.1. Query AI chatbot for Compliance related answers

Description:
This system allows users to query the chatbot for answers related to the compliance documents based on internal policies and regulatory data.

System Input:

- The user enters a compliance question or query through the chatbot's environment.

Display:

- The system exhibits an AI generated answer tailored to the user's question with reference to related policies or regulations.

System Processing:

- The chatbot processes natural language using AI models trained on the compliance information and replies with relevant answers.

System Output:

- Textual replies with citations or links to compliance files are produced to the user.

Other Constraints:

- The system should precisely interpret the user's intention and support confidentiality of important and sensitive data.

Data Handling:

- Deals with natural language input. Replies are generated using AI models with upgradeable compliance knowledge bases.

Error Handling:

- In case the AI model fails to answer, it encourages the user to rephrase or raise the issue to a human expert.

6.8.2. User provide feedback to AI chatbot to improve accuracy over time

Description:

This system allows users to deliver feedback or fixes to AI chatbot replies so the accuracy refines over time

System Input:

- Feedback from users on AI replies which include modifications, additional questions, or confirmations.

Display:

- The system environment displays choices for users to rate, edit, comment or correct the answers.

System Processing:

- Feedback is logged and used to fine-tune the AI models for a seamless improvement.

System Output:

- Recognition and acceptance of feedback and updates reply if required.

Other Constraints:

- Feedback should be safely stored and processed adhering to user anonymity.

Data Handling:

- Logs feedback linked to the question-answer pair for analysis.

Error Handling:

- Manages invalid or inappropriate feedback by alerting, notifying the administrators or by filtering.

6.9.1. Generate Audit Ready Compliance Reports

Description:

Generates structured reports which summarize detected issues and corrective actions

taken and provides the option to download the reports.

System Input:

- Validated compliance documents
- Approved AI suggestions
- Organization metadata such as department, user and timestamps

Display:

- Preview of the audit reports
- Compliance status indicator
- Download Options

System Processing:

1. Collect validated document and approved suggestions

2. Format content into audit ready report

3. Generate visual analytics for dashboard

4. Store report and log actions for audit trail

System Output:

- Downloadable audit ready report
- Dashboard summary
- Audit log entries for tracking

Constraints:

- One validated document is required before report generation
- Report template must be compatible with the selected framework

Data Handling

- Reports must be stored securely in user workspace and archive
- Metadata must be logged for auditing

Error Handling

- If no validation document is available, the system must prompt the user to validate first
- If there is network interruption, report generation will be paused and users will be notified

6.9.2. Maintain Audit Trails and Logs

Description:

The system must automatically record and maintain immutable logs of all user activities and system events related to compliance document handling, including uploads, edits, accesses and AI-generated actions.

System Input:

- User authentication and session data
- Document lifecycle events (upload, modify, delete, view)
- AI processing actions (gap detection, suggestion generation)
- System access attempts (successful and failed)

Display:

- A dedicated 'Audit Logs' section within the admin and compliance officer dashboard.
- Filterable and searchable log entries showing timestamp, user, action and document ID.
- Visual indicators for critical events (eg. Multiple failed access attempts)

System Processing:

1. The system ingests event data from the presentation and logic layers

2. The logic layers “Access Control” and “Collaboration Manager” components timestamp and format the log data.
3. Log entries are written to the secure “Audit Logs Storage” in the Database Layer.
4. Logs are indexed for efficient retrieval and Querying.
5. Logs are cryptographically signed to ensure integrity.

System Output:

- A persistent, tamper-evident record of all auditable events
- Filtered log views and exportable reports (eg. PDF, CSV) for internal reviews or external audits.

Other:

- Logs must be retained for a minimum period as defined by organizational policy or regulatory standards. (at least 3 years to comply with NIST) .
- Access to the full audit trail must be restricted to users with administrative or auditor roles via RBAC.

Constraint:

- Logging mechanisms must not significantly impact system performance or user experience.
- The log story system must be scalable to handle high volumes of events in an enterprise setting.

Data Handling:

- Log data is classified as highly sensitive.
- All log entries are stored in the structured “Audit Logs Storage”
- Data in the audit trail is immutable and cannot be altered or deleted by users.

Error Handling:

- If the log writing process fails, the system must retry according to a defined policy.
- Failed log events must be queued and a system alert must be generated for administrators.
- Users are not interrupted if a non-critical logging action fails.

6.10. Secure Data Storage and Encryption

Description:

The system must ensure that all data, including documents, user information and AI models, is stored securely using robust encryption standards both during transmission and while on disk .

System input:

- User-uploaded compliance documents (PDF, DOCX, etc)
- User credentials and personal data
- System configurations files and application data
- AI model parameters and vector embeddings

Display:

- No direct display of encryption processes,
- The user interface may show security status indicators (e.g. A lock icon, *"Secure Connection")*
- Admin Dashboards may display encryption status reports (e.g. Percentage of encrypted data stores)

System Processing:

- ***For data in transit:***
  - All client-server and inter-service communication is encrypted using TLS/SSL protocols.
- ***For data at rest:***
  - Documents in Cloud Object Storage (AWS S3) are encrypted using AES-256.
  - Database fields containing sensitive data (e.g User details in PostgreSQL) are encrypted.
  - The encryption keys are managed securely, separate from the encrypted data.

System Output:

- Encrypted data stored in persistent storage (S3, databases)
- Secure data streams between system components

Other:

- The system must support the ability to implement client-side encryption for an added layer of security, as required for on-premise deployments.
- Key rotation policies must be defined and enforceable.

Constraints:

- Encryption and decryption processes must be optimized to minimize the latency for the end-user.

- Must comply with data sovereignty laws, meaning encryption keys and data may need to be stored in specific geographic regions.

Data Handling:

- All data is classified based on sensitivity , with compliance documents and user data treated as confidential.
- Data is segmented and stored across different storage solutions (S3 for documents, PostgreSQL for structured data, Vector DB for embeddings) as per the system architecture, with encryption applied to each.

Error Handling:

- If an encryption/decryption operation fails, the system must log the error and fail securely without exposing plaintext data.
- Users must receive a generic error message (e.g. *"Processing error. Please try again")* without technical details.
- Alerts must be sent to system administrators for repeated encryption/decryption failures.

7. External Interface Requirements

7.1 Data Interface

The ComplyAI system exchanges data between users, AI components, and databases to support document analysis, compliance checking, and system learning.

- **Input Data Interfaces**
  - **Document Upload:** Accepts .pdf, .docx or.txt files uploaded by users.
  - **Compliance Standard Feed:** System admins upload or update standards (ISO 27001, GDPR, HIPAA, etc.) to train and refine AI models.
  - **User Input Interface:** Receives user feedback (accept/reject AI suggestions) to improve the system's accuracy.
- **Output Data Interfaces**
  - **Analysis Results:** Provides highlighted non-compliant text, compliance gap reports, and AI confidence scores.

- **Corrected Documents:** Generates downloadable, AI-corrected files maintaining original formatting.
- **Dashboards and Reports:** Displays compliance progress, audit logs, and analytics via web UI.

- **Database Interfaces**
  - **User Database:** Stores credentials, roles, and activity logs.
  - **Compliance Knowledge Base:** Contains compliance standards, guidelines, and AI learning data.
  - **Audit Logs Database:** Records AI actions, document history, and user approvals.
  - **Vector Database:** Stores semantic embeddings for document comparison and retrieval (used in RAG pipeline).
- **Security and Data Flow**
  - Data is encrypted in transit (HTTPS, TLS 1.3) and at rest (AES-256).
  - Temporary processing data is deleted automatically after completion.
  - Access is role-controlled and audited to ensure compliance integrity.

7.2 User Interface

- The system offers a responsive web interface that can be created with React.js or Next.js to be accessible to both desktop and tablet.
- The interface allows users to interact with dashboards, upload panels and compliance analytics visualizations.
- Interfaces will consist of role based access control (Admin, Editor, Viewer) with customized navigation menu.
- The visuals created on Chart.js or D3.js with the help of Chart.js or D3.js are charts, progress indicators, or compliance scores.
- The design is usability compliant, and it provides tooltips, status indicators, and notification panels to provide feedback of the system.
- Users are also able to communicate using a chatbot built-in AI Compliance Copilot to answer immediate questions and get compliance advice.

7.3 Other Interface

- **Communication Interface**:
  - **WebSocket Connections:** Facilitate real-time collaboration, live editing, and progress updates across multiple users.
  - **REST/GraphQL APIs:** Enable communication between the frontend and backend layers for document handling, compliance validation, and AI suggestion retrieval.
  - **Authentication Protocols:** Uses OAuth 2.0 and JWT for secure user sessions, with optional multi-factor authentication for sensitive operations.
  - **Notification Channels:** In-app notifications and email alerts inform users about compliance updates, collaboration invites, and audit readiness status.
- **External System Interfaces:**
  - **Regulatory Data Feeds:** The system can connect with external regulatory sources (e.g.NIST)) to import framework updates and incorporate them into compliance checks.
  - **Audit and Reporting Tools:** Output data and reports can be exported in PDF, DOCX, or CSV formats for integration with external governance and reporting systems.
  - **Security Monitoring Tools:** Optional integration with third-party logging or SIEM tools for compliance tracking and anomaly detection.

8. Non-Functional Requirements

8.1 System Performance

System Performance	
Req Id	Description
PER-1	Response time: 10 seconds for under 50 pages and 30 seconds for upto 200 pages
PER-2	Accuracy of the NLP should at least be 90%
PER-3	The system must be able to process at least 3 documents concurrently without degradation

PER-4	Each identified issue correction must happen within 2 seconds
PER-5	Search queries must be completed within 2-3 seconds for records under 100000 and indexing must be used for faster database queries.

8.2 Information security

Information Security	
Req ID	Description
IS-1	All uploaded compliance documents must be encrypted
IS-2	Role-based access control must prevent unauthorized access
IS-3	Logins must include multi factor authentication
IS-4	System activities must be logged
IS-5	The system must ensure data is stored securely

8.3 Availability

Availability	
Red ID	Description
A-1	The system must be available 99% annually and system downtime for maintenance to not exceed more than 2 hours a week. The system must be available during working hours.
A-2	Daily backups of data must be completed to avoid data loss
A-3	Continuous monitoring of the system is required to detect issues quickly and maintenance must be scheduled in advance with notification sent to users.

8.4 Capacity

Capacity	
Req ID	Description
C-1	Storage capacity of up to 500 GB including uploaded compliance files, reports, and logs.
C-2	Support up to 20 concurrent users and 50 registered accounts
C-3	The organization must be able to manage up to 1000 compliance documents and upload 5 documents simultaneously
C-4	Database must support up to 100000 records of documents and audit logs.